

FORMULAS I

CHAPTER 2: SIMPLE COMPARISONS [2 POPULATIONS]

1. Compare 2 population means (Unknown variances)

Unequal variances (Unknown) Welch t-statistic	$t_0 = \frac{\bar{y}_1 - \bar{y}_2}{\sigma} \sim t_v \quad \text{where } \sigma = \sqrt{\left(\frac{s_1^2}{n_1}\right) + \left(\frac{s_2^2}{n_2}\right)}$ Degree of freedom $v = \frac{(w_1 + w_2)^2}{\frac{w_1^2}{n_1 - 1} + \frac{w_2^2}{n_2 - 1}}; \quad w_1 = \frac{s_1^2}{n_1}, w_2 = \frac{s_2^2}{n_2}$
Equal variances (Unknown): Pooled variances	$t_0 = \frac{\bar{y}_1 - \bar{y}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}.$ (Pooled variance S_p^2)
CI's	$(\bar{y}_1 - \bar{y}_2) \pm t_{\alpha/2, n_1 + n_2 - 2} S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
Paired t-test	$t = \frac{\bar{d}}{\left(\frac{s_d}{\sqrt{n}}\right)} \sim t_{n-1}, \quad d_i = y_{1i} - y_{2i}, \quad i = 1, \dots, n$

2. Distributions- Sampling theory

Theorem	If $X \sim N(\mu, \sigma)$ then $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$
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Chi-square distribution , (n degrees of freedom)	$\chi^2(n) = \sum^n Z_i^2 = Z_1^2 + Z_2^2 + \dots + Z_n^2$ Where $Z_i \sim N(0,1)$
$\chi^2(n)$	If $X \sim \chi^2(n)$ then $E(X) = n; \text{Var}(X) = 2n$ Sum of 2 Chi-square variables is a Chi-square variable $\chi^2(n+m) = \chi_n^2 + \chi_m^2$
Sample Sum Squares SS Population Sum Squares SS_p	SS (Sample Sum Squares) $SS = \sum (x_i - \bar{x})^2$ SS_p (Population Sum Squares) $SS_p = \sum (x_i - \mu)^2$
Chi-square variable related to Sum Squares	$\frac{SS}{\sigma^2} = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1)$

t- distribution, (n degrees of freedom)	$\text{If } Z \sim N(0,1), \quad Y \sim \chi^2(n), \quad \text{then} \quad t = \frac{Z}{\sqrt{\frac{Y}{n}}} = \sim t_n$
	Sample Sum Squares: $\frac{SS}{\sigma^2} = \frac{(n-1)s^2}{\sigma^2} = \chi_{n-1}^2$ $t = \frac{Z}{\left(\frac{s}{\sigma}\right)} = \frac{Z}{\sqrt{\frac{s^2}{\sigma^2}}} = Z / \left(\sqrt{\frac{(n-1)s^2}{\sigma^2(n-1)}} \right) \sim t_{n-1}$ Some result from Basic Statistics [s is used, σ is unknown] $t = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}} = \frac{\sqrt{n}(\bar{X} - \mu)}{s} = \left(\frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} \right) / \left(\frac{s}{\sigma} \right) \sim t_{n-1}$

F- distribution (n_1, n_2 – degrees of freedom)	$F = \frac{\frac{\chi_1^2}{n_1}}{\frac{\chi_2^2}{n_2}} \sim F_{n_1, n_2}$
F statistics	$F_0 = \frac{SS_1}{n_1 - 1} / \left(\frac{SS_2}{n_2 - 1} \right) \sim F_{(n_1 - 1, n_2 - 1)}$
Assume:	$\left(= \left[\frac{(n_1 - 1)s_1^2}{\sigma^2} / (n_1 - 1) \right] / \left[\frac{(n_2 - 1)s_2^2}{\sigma^2} / (n_2 - 1) \right] \right)$
H_0 $\sigma_1^2 = \sigma_2^2 = \sigma$	$= \frac{\frac{\chi_{n_1-1}^2}{n_1 - 1}}{\frac{\chi_{n_2-1}^2}{n_2 - 1}} = F_{(n_1 - 1, n_2 - 1)}$
ANOVA	
Sum Squares	$SST = SSTr + SSE$
Mean Square Total	Sum Square Total = SumSquare Treatment + Sum Square Error
Mean Square Treatment	Mean Square Treatment $MSTr = \frac{SSTr}{df}$
Mean Square Error	Mean Square Error $MSE = \frac{SSE}{df}$

	$F = \frac{MSTr}{MSE} = \frac{\frac{SSTr}{df_{Tr}}}{\frac{SSE}{df_E}} \sim F_{df_{Tr}, df_E}$
Assume: H_0 $\sigma_1^2 = \sigma_2^2 = \sigma$	Assume $\sigma_1 = \sigma_2 = \sigma$ then $F_0 = \frac{s_1^2}{s_2^2} \sim F_{n_1-1, n_2-1}$ (Some result from Basic Statistics)
Regression-Mean Squares (ANOVA F-test) Ho: $\beta_0 = \beta_1 = \dots \beta_a = 0$	$SST = SSR + SSE$ (Sum Square Total = Sum Square Regression + Sum Square Error) $MSR = SSR/df_R$; $MSE = SSE/df_E$ $F_0 = \frac{MSR}{MSE} \sim F_{(k, n-k-1)}, k \text{ regressors}$
t and F	$t^2 = \frac{z^2}{\left(\frac{s^2}{\sigma^2}\right)} = \left(\frac{\chi^2}{1}\right) / \frac{(n-1)s^2}{\sigma^2(n-1)} = F_{1, n-1}$

CHAPTER 3: 1- FACTOR ANALYSIS[One factor with a number of treatment levels]

Tr 1	y_{11}	y_{12}	..	y_{1n_1}
Tr 2	y_{21}	y_{22}	..	y_{2n_2}
...				
Tr a	y_{a1}	y_{a2}	..	y_{an_a}

Suppose there are : a treatment level (or factor level) , n_i replicates $i = 1, \dots, a$

Balanced design if, $n_1 = n_2 = \dots = n_a = n$
 Unbalanced if otherwise

Random Effects Model: $y_{ij} = \mu + \tau_i + \epsilon_{ij}$ $i = 1, \dots, a$, treatment levels $j = 1, \dots, n_i$, number of replicates $\tau_i \sim N(0, \sigma_\tau^2)$, $\epsilon_{ij} \sim iid N(0, \sigma_\epsilon^2)$,	$\tau_i \sim N(0, \sigma_\tau^2)$ Parameters to be estimated: $\mu, \sigma_\tau^2, \sigma_\epsilon^2$
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Fixed Effects models: $y_{ij} = \mu + \tau_i + \epsilon_{ij}$, $\tau_i, i = 1, \dots, a$ are fixed $\epsilon_{ij} \sim iid N(0, \sigma_e^2)$ Parameters: μ, τ_i (a of them), σ_e^2	$\tau_i, i = 1, \dots, a$ are fixed. Assume: $\sum_{i=1}^k \tau_i = 0 .$ Parameters to be estimated : μ, τ_i (a of them), σ_e^2
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Fixed Effects Models (another way) $y_{ij} = \mu_i + \epsilon_{ij}$, $i = 1, \dots, a \quad j = 1, \dots, n_i \quad \epsilon_{ij} \sim iid N(0, \sigma^2)$	Parameters to be estimated: μ_i, σ^2
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Treatment Effects $y_{ij} = \bar{y}_{..} + (\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})$ Or: $y_{ij} - \bar{y}_{..} = (\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})$ effects errors	Treatment effects: $i^{th} - effect: \tau_i = \bar{y}_{i.} - \bar{y}_{..}$ $\sum_{i=1}^k \tau_i = 0 .$ Assume: $ij^{th} - residuals \quad \epsilon_{ij} = y_{ij} - \bar{y}_{i.}$
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Hypothesis Testing: Test Equality of Treatment means $H_0: \mu_1 = \mu_2 = \dots = \mu_a = \mu$, or $H_0: \tau_1 = \tau_2 = \dots = \tau_a = 0$ $H_A: \mu_i \neq \mu_j$ $H_A: \tau_i \neq 0$, for some i

ANOVA – F-test $\sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{..})^2 = \sum_{i=1}^k n_i (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_{i.})^2$ Sum Square Total = Sum Square Treatments + Sum Square Error (SST) SSTr Where, $\bar{y}_{i.} = \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij}, \quad \bar{y}_{..} = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} y_{ij}$
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SSTreatment (or SS Between),
SS Error (or SS Within)

$$MSTr \text{ (or MSB)} = \frac{SSTr}{a-1} \quad ; \quad MSE \text{ (or MSW)} = \frac{SSE}{N-a} ,$$

$$F_0 = MSTr/MSE \sim F_{a-1, N-a}$$

ANOVA	Source	SS	df	MS	Fo
	Treatmen (or Between)	SSTr	$a-1$	MSTr	MSTr/MSE
	Error (or Within)	SSE	$N-a$	MSE	
	Total	SST	$N-1$		

1. Parameters estimation (Point estimate / Interval estimate)

$y_{ij} = \bar{y}_{..} + (\bar{y}_{i.} - \bar{y}_{..}) + (y_{ij} - \bar{y}_{i.})$	Effects and error equation
Effects ($\tau_i, i = 1, \dots, a$) $\hat{\tau}_1 = \bar{y}_{1.} - \bar{y}_{..}$ $\hat{\tau}_2 = \bar{y}_{2.} - \bar{y}_{..}$ Residuals $ij^{th} - \text{residual } \epsilon_{ij} = y_{ij} - \bar{y}_{i.}$	

CI's for μ_i's $\bar{y}_{i.}, i =$ $1, \dots, a$, are unbiased estimates of μ_i 's	$\bar{y}_{i.} \pm t_{(\frac{\alpha}{2}, N-a)} \sqrt{\frac{MSE}{n_i}}$
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CI's for Difference between 2 treatment means $(\mu_i - \mu_j)$	$(\bar{y}_{i.} - \bar{y}_{j.}) \pm t_{\left(\frac{\alpha}{2}, N-a\right)} \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$
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(a) Fisher Pairwise t-test

Fisher Pairwise t-Test/ CI's (Multiple Pairwise Comparison) $H_0: \mu_1 = \mu_2 = \dots = \mu_a$ $H_a: \mu_i \neq \mu_j$	$t_0 = (\bar{y}_{i.} - \bar{y}_{j.}) / \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$ compare with $t_{\frac{\alpha}{2}, N-a}$ OR: compare $ \bar{y}_{i.} - \bar{y}_{j.} $ with Fisher LSD Fisher LSD $LSD = t_{\left(\frac{\alpha}{2}, N-a\right)} \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$ $LSD = t_{\left(\frac{\alpha}{2}, N-a\right)} \sqrt{\frac{2MSE}{n}} \text{ for balanced data}$
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(b) Contrasts

Contrast	A contrast is any linear combination of the population means $C = c_1\mu_1 + c_2\mu_2 + \dots + c_a\mu_a, \quad \sum c_i = 0$ Example: $C = \mu_i - \mu_j$ (compare pair of mean) $C = \mu_1 - 2\mu_2 + \mu_3$	
Test a contrast Ho: $C = 0$ Ha: $C \neq 0$ Balanced data	$t_o = \frac{C}{se(C)}$ $\sim t_{(N-a)}$ $C = c_1\bar{y}_{1.} + c_2\bar{y}_{2.} + \dots + c_a\bar{y}_{a.}$ $var(C) = \frac{\sigma^2}{n} (\sum c_i^2) = \frac{MSE}{n} (\sum c_i^2)$ $se(C) = \sqrt{\frac{MSE}{n} (\sum c_i^2)}$ (MSE replaces σ^2). For simple contrast, it becomes the Fisher LSD $t_0 < t_{\frac{\alpha}{2}, N-a} \Rightarrow C = 0$; $t_0 > t_{\frac{\alpha}{2}, N-a} \Rightarrow C \neq 0$	
Unbalanced data	$var(T) = MSE \left(\sum \frac{c_i^2}{n_i} \right)$	

(c) Other Pairwise Comparison Test

Scheffe (compare any contrast)	Scheffe critical value $S_{\alpha,u} = se(C) \sqrt{(a-1)F_{\alpha,a-1,N-a}}$ If $ C < S_{\alpha} \Rightarrow C = 0$; $ C > S_{\alpha} \Rightarrow C \neq 0$
Tukey Compare contrast $C = \mu_i - \mu_j$ (compare pairwise) Tukey distribution stat	$C_u = \mu_i - \mu_j$ To test: $H_0: \mu_i = \mu_j$ Tukey critical value , against which $ \bar{y}_i - \bar{y}_j $ compare $T_{\alpha} = q(\alpha, f) \sqrt{\frac{MSE}{n}}$ f : degree of freedom of MSE

(d) Power-Sample size

Sample size	Type II error: $\beta = P(\text{Accept } H_0 H_0 \text{ is False})$ Power of a test: $\text{Power} = 1 - \beta = 1 - P(\text{Accept } H_0 H_0 \text{ False})$ $= P(\text{Reject } H_0 H_0 \text{ False})$ $= P(F_0 > F_{\alpha,a-1,N-a} H_0 \text{ False})$ $= P(\text{Reject } H_0 H_0 \text{ False})$
SStr, SSE	a) $\frac{SStr}{\sigma^2} \sim \chi_{a-1}^2$ b) $\frac{SSE}{\sigma^2} \sim \chi_{N-a}^2$ c) $F_0 = \frac{MStr}{MSE} \sim F_{a-1,N-a}$
More on variances	a) $E(MSE) = \sigma^2$ b) $E(MStr) = \sigma^2 + n \sigma_{\tau}^2 = \sigma^2 + \frac{n(\sum \tau_i^2)}{a-1}$ <i>Estimators:</i> a) $\sigma^2 = MSE$ b) $\sigma_{\tau}^2 = \frac{MStr - MSE}{n}$

Balanced and Unbalanced data	a) $SSTr = \Sigma^a n_i \tau_i^2; \quad MSTr = \frac{SSTr}{a-1}$ (Unbalanced) b) $SSTr = \Sigma^a n \tau_i^2 = n \Sigma^a \tau_i^2; MSTr = n \frac{\Sigma^a \tau_i^2}{a-1}$ (Balanced)
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Example

The tensile strength of a synthetic fiber used to make cloth for men's shirts is of interest to a manufacturer. It is suspected that the strength is affected by the percentage of cotton in the fiber. Five levels of cotton percentage are considered: 15%, 20%, 25%, 30% and 35%. For each percentage of cotton in the fiber, strength measurements (time to break when subject to a stress) are made on five pieces of fiber.

15	20	25	30	35
7	12	14	19	7
7	17	18	25	10
15	12	18	22	11
11	18	19	19	15
9	18	19	23	11

The corresponding ANOVA table is

Source	df	SS	MS	F_0
Between	4	475.76	118.94	$F_0 = 14.76$
Within (Error)	20	161.20	8.06	
Total	24	636.96		

Performing the test at $\alpha = .01$ one can easily conclude that the percentage of cotton has a significant effect on fiber strength since $F_0 = 14.76$ is greater than the tabulated $F_{4,20}(.01) = 4.43$.

The estimate of the overall mean is $\hat{\mu} = \bar{y}_{..} = 15.04$. Point estimates of the treatment effects are

$$\begin{aligned}
\hat{\tau}_1 &= \bar{y}_{1.} - \bar{y}_{..} = 9.80 - 15.04 = -5.24 \\
\hat{\tau}_2 &= \bar{y}_{2.} - \bar{y}_{..} = 15.40 - 15.04 = 0.36 \\
\hat{\tau}_3 &= \bar{y}_{3.} - \bar{y}_{..} = 17.60 - 15.04 = -2.56 \\
\hat{\tau}_4 &= \bar{y}_{4.} - \bar{y}_{..} = 21.60 - 15.04 = 6.56 \\
\hat{\tau}_5 &= \bar{y}_{5.} - \bar{y}_{..} = 10.80 - 15.04 = -4.24
\end{aligned}$$

A 95% percent CI on the mean treatment 4 is

$$21.60 \pm (2.086)\sqrt{8.06/5},$$

which gives the interval $18.95 \leq \mu_4 \leq 24.25$.

Example

Consider the fabric strength example we considered above. The ANOVA F -test rejected $H_0 : \mu_1 = \dots = \mu_5$. The LSD at $\alpha = .05$ is

$$\text{LSD} = t_{20}(.025) \sqrt{MS_W(1/5 + 1/5)} = 2.086 \sqrt{\frac{2(8.06)}{5}} = 3.75 .$$

Thus any pair of treatment averages that differ by more than 3.75 would imply that the corresponding pair of population means are significantly different. The $\binom{5}{2} = 10$ pairwise differences among the treatment means are

$\bar{y}_{1.} - \bar{y}_{2.} =$	$9.8 - 15.4 =$	-5.6^*
$\bar{y}_{1.} - \bar{y}_{3.} =$	$9.8 - 17.6 =$	-7.8^*
$\bar{y}_{1.} - \bar{y}_{4.} =$	$9.8 - 21.6 =$	-11.8^*
$\bar{y}_{1.} - \bar{y}_{5.} =$	$9.8 - 10.8 =$	-1.0
$\bar{y}_{2.} - \bar{y}_{3.} =$	$15.4 - 17.6 =$	-2.2
$\bar{y}_{2.} - \bar{y}_{4.} =$	$15.4 - 21.6 =$	-6.2^*
$\bar{y}_{2.} - \bar{y}_{5.} =$	$15.4 - 10.8 =$	4.6^*
$\bar{y}_{3.} - \bar{y}_{4.} =$	$17.6 - 21.6 =$	-4.0^*
$\bar{y}_{3.} - \bar{y}_{5.} =$	$17.6 - 10.8 =$	6.8^*
$\bar{y}_{4.} - \bar{y}_{5.} =$	$21.6 - 10.8 =$	10.8^*

Using underlining the result may be summarized as

$\bar{y}_{1.}$	$\bar{y}_{5.}$	$\bar{y}_{2.}$	$\bar{y}_{3.}$	$\bar{y}_{4.}$
<u>9.8</u>	<u>10.8</u>	<u>15.4</u>	<u>17.6</u>	21.6

As k gets large the experimentwise error becomes large. Sometimes we also find that the LSD fails to find any significant pairwise differences while the F -test declares significance. This is due to the fact that the ANOVA F -test considers all possible comparisons, not just pairwise comparisons.

We start with the Bonferroni Inequality. Let $A_1, A_2, \dots, A_k$ be k arbitrary events with $P(A_i) \geq 1 - \alpha/k$. Then $P(A_1 \cap A_2 \cap \dots \cap A_k) \geq 1 - \alpha$.

The proof of this result is left as an exercise.

We may use this inequality to make simultaneous inference about linear combinations of treatment means in a one-way fixed effects ANOVA set up.

Let $L_1, L_2, \dots, L_r$ be r linear combinations of $\mu_1, \dots, \mu_k$ where $L_i = \sum_{j=1}^k l_{ij} \mu_j$ and $\hat{L}_i = \sum_{j=1}^k l_{ij} \bar{y}_j$ for $i = 1, \dots, r$.

A $(1 - \alpha)100\%$ simultaneous confidence interval for $L_1, \dots, L_r$ is

$$\hat{L}_i \pm t_{n-k} \left(\frac{\alpha}{2r} \right) \sqrt{MS_W \sum_{j=1}^k l_{ij}^2 / n_j}.$$

for $i = 1, \dots, r$.

A Bonferroni α -level test of

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

is performed by testing

$$H_0 : \mu_i = \mu_j \quad \text{vs.} \quad H_A : \mu_i \neq \mu_j$$

with

$$t_{ij} \frac{|\bar{y}_i - \bar{y}_j|}{\sqrt{MS_W(1/n_i + 1/n_j)}} > t_{n-k} \left(\frac{\alpha}{2 \binom{k}{2}} \right),$$

for $1 \leq i < j \leq k$.

There is no need to perform an overall F -test.